PHYSICS I / COURSE MATERIAL

Newton's Laws of Motion

PlusOne teaching sample | first-year university | 15-20 min reading

Purpose. Describe motion using forces, and defend explanations with peers. This document is original demonstration content aligned to OpenStax *University Physics*, Volume 1, Chapter 5. It is not an official Polito course handout.

Learning goals

- **LO1.** Distinguish velocity from acceleration and apply Newton's first law.
- **LO2.** Compute acceleration from the vector sum of external forces and mass.
- **LO3.** Identify Newton's third-law force pairs on different interacting bodies.
- **LO4.** Construct a free-body diagram with only forces acting on the chosen body.
- **LO5.** Reason about static and kinetic friction without assuming friction has a fixed value.
- **LO6.** Explain reasoning to peers and apply a principle to a new situation.

A. Start with the system, not the equation

A **system** is the body or collection of bodies you choose to analyze. A **force** is an interaction: a push, pull, support, friction or gravitational attraction. Forces are vectors. The **net external force** is the vector sum of external forces acting on the chosen system.

For a free-body diagram (FBD): isolate one body; draw only forces **on** it; label the source of each; choose axes; add components; apply Newton's second law. Do not add the acceleration as an extra force.

B. The three laws

First law (inertia). In an approximately inertial reference frame, a body with zero net force keeps its velocity unchanged, including the case of zero velocity. Force changes velocity; force is not required to maintain constant velocity.

Second law. The vector sum of external forces equals mass times acceleration: **$\Sigma F = ma$**. In one dimension, choose a positive direction and preserve signs. An acceleration to the right may coexist with motion to the left.

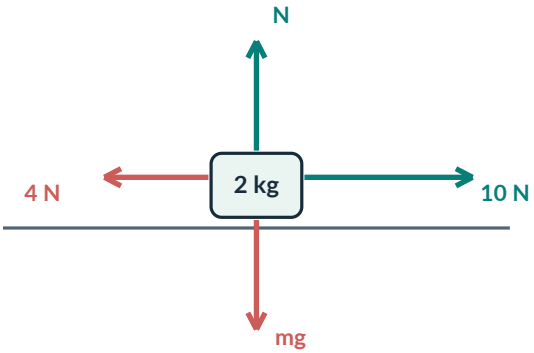
Third law. If A exerts a force on B, B simultaneously exerts an equal-magnitude, opposite-direction force on A. The two forces act on **different bodies**, so they do not cancel on a single body's FBD.

Key distinction

Balanced forces: same body, vector sum can be zero. **Third-law pair:** different bodies in one interaction. Do not confuse these two ideas.

C. Draw and reason with forces

Worked example: a horizontal cart. A 2.0 kg cart is pulled right with 10 N and experiences a 4 N opposing horizontal force. The vertical forces cancel. What is its acceleration?



Solution: Taking right as positive, $\Sigma F_x = +10\text{ N} - 4\text{ N} = +6\text{ N}$. Therefore $a_x = (6\text{ N})/(2.0\text{ kg}) =$ **3.0 m/s² to the right**. The direction and units are part of the answer.

Notice: The cart could be moving either left or right at that instant. Its acceleration is still to the right because that is the direction of the net force.

D. What an FBD must not do



Equal and opposite interaction forces act on different bodies.

If A pushes B right, the interaction reaction is B pushing A left. They belong on separate FBDs. By contrast, for a book resting on a table, the upward normal force and downward weight are forces *on the same book*; they can balance but they are not a third-law pair.

E. Friction, inclines, and common traps

- **Static friction** adjusts to what is needed to prevent slipping, up to a limiting maximum. If a 12 N push does not move a box, the friction force balances it at 12 N in the opposite direction, assuming no other horizontal force.
- **Kinetic friction** opposes relative sliding between contacting surfaces, not necessarily the direction of acceleration.
- **Inclined plane:** weight has a component $mg \sin\theta$ parallel down the slope and $mg \cos\theta$ perpendicular to the slope. For a frictionless incline, acceleration along the slope is $g \sin\theta$.
- **Heavy truck versus small car:** collision interaction forces are equal and opposite. Their accelerations differ because their masses differ.

F. Group discussion protocol

Each student first writes an independent prediction. During discussion: state a claim, name the force(s) on the chosen body, ask a peer to challenge the claim, and revise if necessary. PlusOne should ask clarifying questions and record evidence, not supply the final reasoning.

Useful self-check questions

- What object or system am I analyzing?
- Which forces are external to it?
- Does the net force imply acceleration, or does it imply constant velocity?
- Do my alleged third-law forces act on different bodies?

Sources and attribution

OS-5.1: OpenStax University Physics Vol. 1, Section 5.1 - Forces. https://openstax.org/books/university-physics-volume-1/pages/5-1-forces

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